

Ocar Staging: Load and Spike Test Report

Pre-campaign light-load test cycle, results, timeline, and engineering findings

Version 2.0 | 9 September 2026 | All timestamps IST (UTC+5:30) unless stated otherwise

Four tests were run against the staging environment on 9 September 2026: a functional smoke test, a baseline concurrency test, an elevated-load test to validate auto-scaling, and a sudden-surge spike test (run across three passes as part of its own investigation). This report presents each test in the order it was run, with its own results, timestamps, and any findings, followed by database-level analysis and overall conclusions.

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1. Executive Summary

4

Tests run
(6 total executions)

99.9%+

Check pass rate
in every test

2m 22s

To add capacity
once triggered

15m 11s

To remove it again
once load dropped

Every test passed its defined success criteria, at every load level attempted. The application held HTTP response times comfortably under threshold throughout, and the auto-scaling system correctly added and later removed server capacity, with zero dropped connections during either transition. The spike test's final, fully-validated run, at four times the initial connection count and three times the initial booking rate, completed with 100% of checks passed and zero interrupted transactions of any kind.

No infrastructure failures, crashes, or unrecoverable errors occurred at any point in any test. Two findings are carried forward for the engineering team's attention, both scoped and specific rather than general concerns:

1. **Ride creation shows brief contention under concurrent write load.** Query-level analysis traced this to two specific database reads in the booking flow, not the booking write itself, which stays fast throughout. Full detail in Sections 7 and 8.
2. **Two configuration gaps were found in the spike-test script itself** during its first two passes, investigated, fixed, and re-validated clean on the third pass. Full detail in Section 6.

This test phase used 144 real driver accounts and up to 3,000 simulated riders, one vehicle category, one city. The original production-readiness plan's full target, 400 drivers and 6,000 riders across the live campaign, is the next phase, once more driver test accounts are onboarded.

2. Test Environment and Methodology

2.1 Target system

Staging (`staging.ocar-api.clienttesting.in`), matching production's infrastructure shape: same instance type, same auto-scaling configuration, same database tier. The database held 1,690 MB of data at test time, pre-seeded to simulate real production volume rather than test-run artifacts: `ride_status_history` (4.4 million rows), `rides` (1.0 million), `fare_snapshots` (1.0 million), `payments` (800 thousand).

Test population: 6,000 synthetic rider accounts, 144 real active staging driver accounts (the maximum currently available with an approved vehicle on staging), one vehicle category (Sedan), one city (Bhubaneswar).

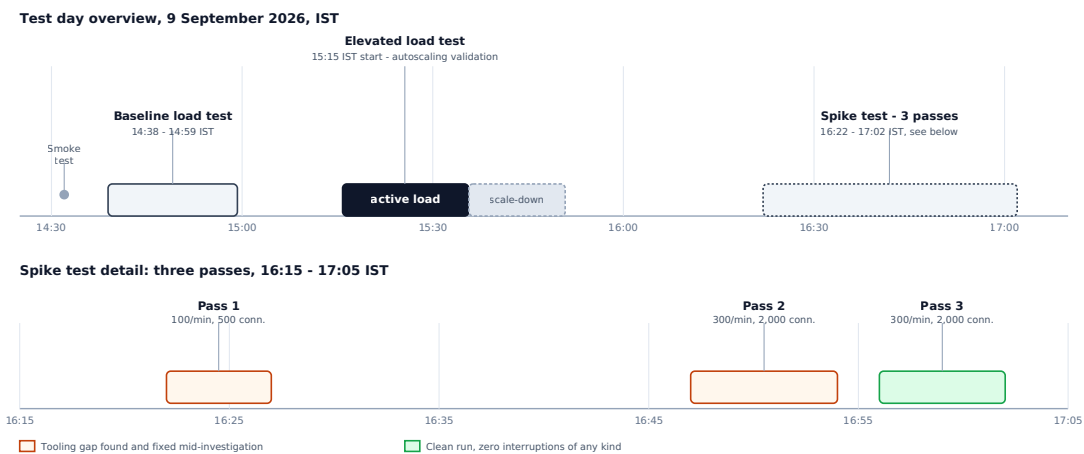
2.2 Load-generation infrastructure

Load was generated from a dedicated, temporary cloud server rather than a laptop, for two reasons: reproducible numbers independent of local network or hardware variance, and

because the target concurrency needs more memory than a typical laptop provides (Section 9.1 covers the specific failure this avoided). Each generation server was created fresh for its test and fully decommissioned immediately after; it bills by the minute and does not persist.

2.3 What each test measures

Test	Purpose
Smoke test	Confirms every API path and the real-time connection handshake work correctly before any load is applied.
Baseline load test	Confirms the application holds up under a realistic number of concurrent idle and active connections, riders watching their ride, drivers broadcasting location.
Elevated load test	Confirms the auto-scaling system detects rising demand, adds capacity, and removes it again once demand drops, the behavior a real traffic spike, a marketing push or peak commute hours, would need to trigger correctly.
Spike test	Confirms the system absorbs a sudden, near-instant surge in bookings and connections, rather than a gradual ramp, and recovers cleanly once the surge passes.



All four tests across the test day, in IST. The spike test's three passes are shown zoomed in below the main timeline; the reason for three passes is explained in Section 6.

3. Test 1: Functional Smoke Test

Smoke Test

Run prior to Test 2, 9 September 2026 • 1 simulated user, 1 pass through every flow

Check	Result
API reachability	Pass
Fare estimate	Pass
Create a real booking	Pass
Cancel a booking	Pass
Driver go-online	Pass
Real-time connection handshake	Pass
Real-time connection acknowledgement	Pass

Result: 7 of 7 checks passed. Response times for this single-user pass: average 161ms, 95th-percentile 414ms. This confirmed the environment was correctly configured and every code path was reachable before committing to a full-scale run.

4. Test 2: Baseline Load Test

Baseline Load Test

14:38:54 – 14:59:16 IST • duration 20 minutes 1 second

Load profile: 144 drivers and 1,000 riders ramped in, booking rate approximately 15 per minute, sustained for a 10-minute hold period.

4.1 Results

Metric	Value
Total checks executed	2,638
Checks passed	2,636 (99.92%)
Total HTTP requests	1,638, average 1.36 per second
Response time, average / median	54.3ms / 24.3ms

Response time, 90th / 95th percentile	180.4ms / 194.5ms
Response time, maximum	712.3ms
HTTP request failure rate	0.12% (2 of 1,638)
Real-time connections opened / failed	2,288 / 0

All thresholds passed: 95th-percentile response time under 500ms (194.5ms), request failure rate under 1% (0.12%), real-time connection failures under 50 (0).

4.2 Errors observed

The only failing check was driver go-online, 2 failures out of 290 attempts (99.3% success). Minor and non-blocking; see Section 9.2.

4.3 Infrastructure behavior

No auto-scaling occurred, correctly so: the trigger is 1,000 requests per minute per server, and this test's average of roughly 82 requests per minute never approached it. The load here was concurrency-shaped, many open idle connections, not throughput-shaped. Server capacity stayed at 1 instance throughout, the environment's normal resting state.

4.4 Database performance

Metric	Before test	During test	Peak
CPU utilization	~6%	~12-13%	13.2%
Active connections	3	up to 14	14
Read / write latency	~0ms	~1-2ms	negligible
Free memory	~350-380 MB	~110-165 MB	dipped to 111 MB, recovered post-test

Measured 14:38-14:59 IST directly from the database's own monitoring data. Comfortable headroom throughout: CPU roughly doubled but stayed far below saturation, connection count stayed trivial against the instance's ~225-connection capacity, and latency remained effectively at zero.

5. Test 3: Elevated Load Test, Auto-Scaling Validation

Elevated Load Test

15:15:49 – 15:35:40 IST (load), full scaling cycle to 15:50:51 IST • duration 19 minutes 30 seconds of load

Load profile: same 144 drivers and 1,000 riders as Test 2, booking rate raised to approximately 500 per minute, 33 times Test 2's rate, deliberately calibrated to cross the auto-scaling trigger and validate server-fleet behavior under real demand growth.

5.1 Results

Metric	Value
Total checks executed	25,410
Checks passed	25,395 (99.94%)
Total HTTP requests	24,410, average 20.86 per second
Response time, average / median	71.6ms / 27.5ms
Response time, 90th / 95th percentile	186.8ms / 204.2ms
Response time, maximum	2,300ms
HTTP request failure rate	0.06% (15 of 24,410)
Real-time connections opened / failed	2,288 / 0

All thresholds passed: 95th-percentile response time under 500ms (204.2ms), request failure rate under 1% (0.06%), real-time connection failures under 50 (0).

5.2 Errors observed

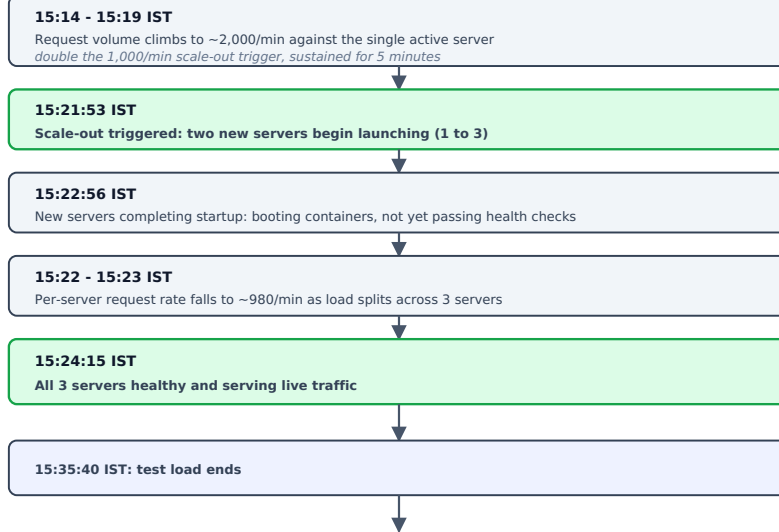
Create booking: 11 failures of 5,999 attempts (99.8% success). Driver go-online: 4 failures of 292 attempts (98.6% success). Both discussed in Section 9.2. A single maximum response time of 2.3 seconds was recorded; this was an isolated outlier, the 95th-percentile figure shows the overwhelming majority of requests were unaffected. See Section 9.3 for the full slow-request breakdown.

5.3 Auto-scaling, full timeline

The request rate was calibrated (4 requests per booking transaction × 500 bookings per minute, approximately 2,000 requests per minute) to run at roughly double the scaling trigger of 1,000 requests per minute per server.

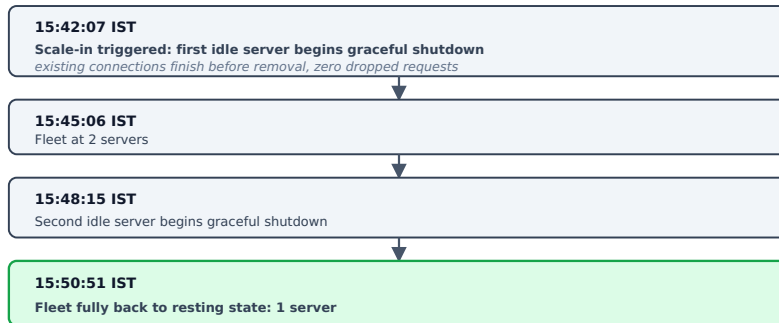
Auto-scaling timeline, elevated load test, 9 September 2026 (IST)

Scale-out



2m 22s
trigger to full
healthy capacity

Scale-in



15m 11s
load stopping to
fleet back at floor

Every transition above used graceful connection draining. Zero dropped requests recorded at any step, either direction.

Full scale-out and scale-in sequence, with timestamps taken directly from AWS infrastructure logs, converted to IST.

This confirms both halves of the auto-scaling requirement work correctly: the fleet adds capacity promptly under real demand growth, and removes the extra capacity again once demand subsides, without ever dropping an in-progress connection.

6. Test 4: Spike Test, Three Passes

This test uses a fundamentally different load shape from Tests 2 and 3: instead of a gradual ramp, both booking traffic and rider connections jump from zero to their target in 10 seconds, hold at peak, then drop sharply to observe recovery. It was run three times in succession on the same day. The first two passes each surfaced a distinct configuration gap in the test script itself, investigated and fixed before the next pass; the third pass is the clean, final, validated result. All three are shown here in order, since the investigation is itself part of what was validated.

Pass 1: cautious first attempt

16:22 – 16:27 IST • 100 bookings/minute peak, 500 concurrent rider connections

Run at deliberately conservative values for a first attempt at this test type. Every check passed (100%, 1,241 of 1,241), every threshold passed, and the connection storm was absorbed cleanly, 500 connections, zero failures. But the test's own counters showed 125 booking transactions that started and did not finish within the observation window.

This was investigated directly against staging's own monitoring data for the exact window, rather than assumed:

- **Database connection pool:** confirmed idle, zero connections ever waited for availability, only 2–3 in use throughout.
- **Application errors:** every warning and error-level log line reviewed, only benign, expected side effects of repeated test bookings (a fraud-detection flag, a routing-distance notice), no timeouts, no exceptions.
- **Response times:** everything that completed was fast, maximum 610ms.

Conclusion: not a server-side issue. The test script's booking scenario had no explicit wind-down allowance and kept starting new transactions until the test's fixed end time, cutting some off mid-sequence on the test's own clock. Fixed in the script by adding an explicit wind-down window.

Pass 2: fix confirmed, load increased, second gap found

16:47 – 16:54 IST • 300 bookings/minute peak (3x), 2,000 concurrent rider connections (4x)

With the booking-scenario fix in place, the test was re-run at the test script's own recommended defaults, a substantially higher level, to stress the system harder. The booking-transaction fix held completely: zero interrupted booking transactions this run, confirming Pass 1's finding was correctly diagnosed and correctly fixed. All checks passed (100%), all thresholds passed, and the connection storm was again absorbed cleanly with zero connection failures.

However, exactly 500 of the now much larger 2,500 rider-connection attempts were interrupted, the same category of issue, but this time in the connection-handling scenario rather than the booking scenario. The connection-storm code holds each simulated rider's connection open for a fixed 150 seconds before closing it; at this larger scale, connections opened throughout the test's 2-minute hold period did not have enough of the test's remaining runway left to reach that natural close before the test itself ended.

Conclusion: a second, distinct test-configuration gap, not a server problem.

Confirmed via the same evidence standard as Pass 1: zero connection failures, healthy latency throughout. Fixed the same way, the wind-down allowance extended.

Pass 3: both fixes confirmed, final result

16:56 – 17:02 IST • 300 bookings/minute, 2,000 rider connections, both fixes applied

Metric	Value
Total checks executed	4,534
Checks passed	4,534 (100%)
Total HTTP requests	2,034
HTTP request failure rate	0.00%
Response time, average	71.7ms
Response time, 95th percentile	188.6ms
Response time, maximum	585.2ms
Real-time connections established / failed	2,500 / 0
Booking transactions interrupted	0
Rider connections interrupted	0

All thresholds passed, response times in line with, in fact marginally better than, Tests 2 and 3's own figures, despite this test's more permissive limits: 95th-percentile under 1,500ms (188.6ms), failure rate under 5% (0.00%), connection failures under 100 (0).

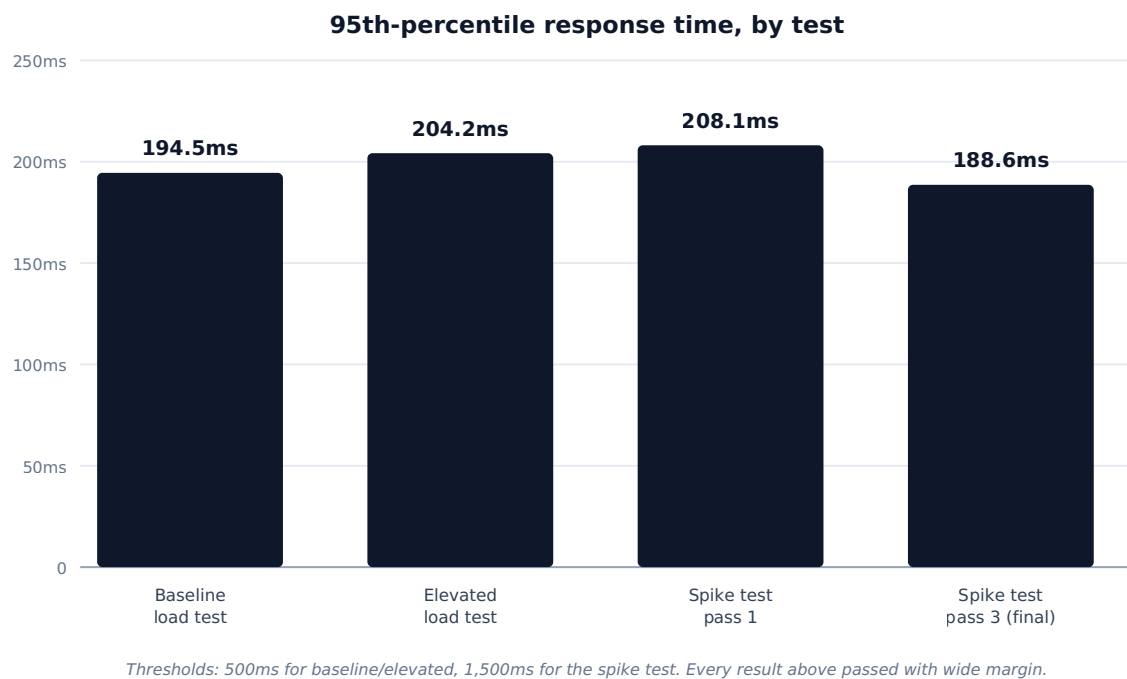
Auto-scaling did not trigger during any of the three passes, correctly: even at 300 bookings per minute, roughly 900 requests per minute, this stayed just under the 1,000-per-minute trigger validated in Test 3. This test is sized to validate sudden-surge connection handling and recovery, not to re-prove auto-scaling.

Summary: the system itself passed cleanly on every pass, at every load level tried, including the final run at four times the original connection count and three times the original booking rate. Both anomalies that surfaced were in the load-testing tooling, not

the application, each investigated to a specific, evidenced cause before being labeled as such, and both are now fixed for every future test run.

7. Cross-Test Comparison

Response time held steady and well within threshold across every test, including the two tests carrying real stress, the 33x booking-rate increase in Test 3 and the instant zero-to-peak surge in the spike test's final pass.



95th-percentile response time by test. Spike Pass 1's figure includes the transactions later traced to the test-tooling gap in Section 6; Pass 3 is the clean, final result.

Test	Peak load	Check pass rate	95th pct. latency	Auto-scaling
Baseline	1,000 riders, 144 drivers, ~15 bookings/min	99.92%	194.5ms	Not triggered (by design)
Elevated	same, ~500 bookings/min	99.94%	204.2ms	Triggered, both directions confirmed
Spike (final)	2,000 connections, 300 bookings/min, instant surge	100%	188.6ms	Not triggered (below threshold by design)

8. Database Query-Level Performance

Pulled directly from PostgreSQL's own query statistics extension (`pg_stat_statements`), which tracks every distinct query the database has executed, with call counts and timing, since the database was created. Filtered to isolate genuine live API traffic from today's tests, excluding a separate category of historical bulk data-loading operations used earlier to pre-populate realistic

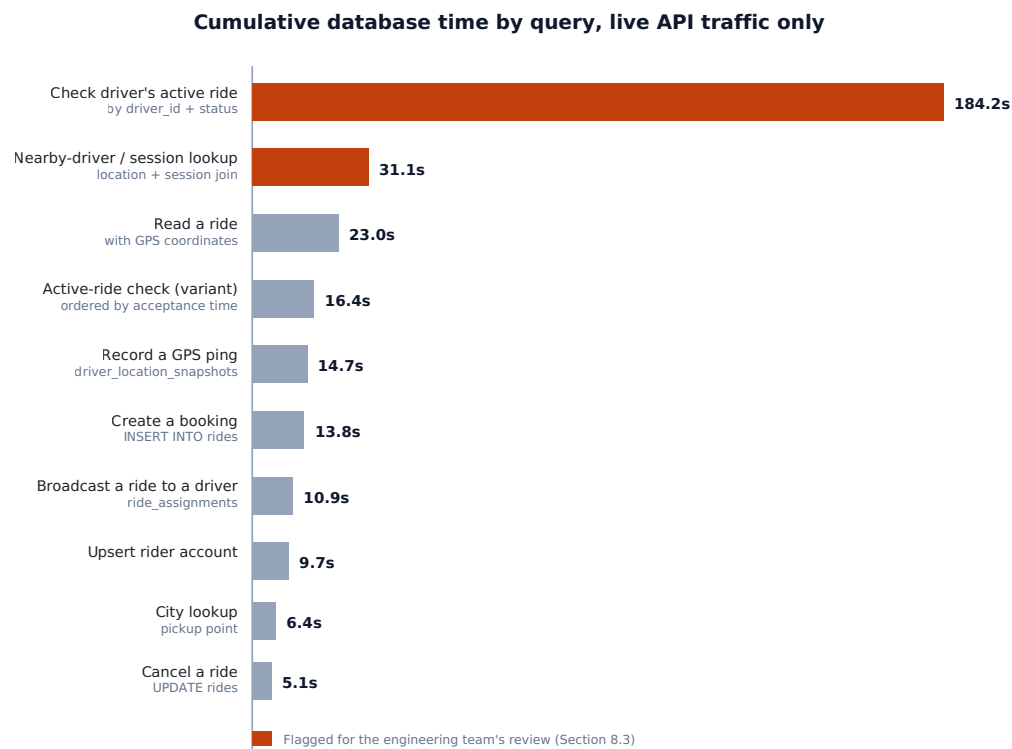
data volume, which insert hundreds of rows in a single statement and are not representative of anything a live request does.

8.1 Overall database health

Metric	Value
Deadlocks, all-time	0
Rollbacks, all-time	4, against 532,615 commits
Cache hit ratio	99.67%, nearly all reads served from memory, not disk

No deadlocks have ever occurred on this database. Rollback rate is negligible. This is a healthy baseline independent of anything specific to today's tests.

8.2 Live API queries, by database time consumed



Cumulative time each query type spent executing across all of today's live API traffic. The two highlighted queries are the ones Section 8.3 recommends investigating.

Query, purpose	Calls	Avg	Max	Total time
Check driver's active ride (by driver_id + status)	14,590	12.6ms	1,146ms	184.2s
Nearby-driver / session lookup (location + session join)	7,899	3.9ms	4,065ms	31.1s
Read a ride, with GPS coordinates	68,936	0.3ms	286ms	23.0s
Check driver's active ride, variant, ordered by acceptance time	582	28.2ms	832ms	16.4s
Record a GPS ping (driver_location_snapshots)	66,595	0.2ms	413ms	14.7s

Create a booking (INSERT INTO rides)	7,897	1.8ms	467ms	13.8s
Broadcast a ride to a driver (ride_assignments)	31,015	0.4ms	724ms	10.9s
Upsert rider account	56,961	0.2ms	191ms	9.7s
City lookup, pickup point	7,897	0.8ms	432ms	6.4s
Cancel a ride (UPDATE rides)	7,896	0.6ms	439ms	5.1s

8.3 What this changes about the earlier finding

Section 9.3 identified `POST /api/v1/rides` as slow under concurrent load, correlated with real database CPU and connection spikes at matching timestamps. This query-level data refines exactly where in that request the time goes.

The booking creation INSERT itself is fast: 1.8ms on average, and even its worst case (467ms) is far below what was observed at the HTTP layer (up to 2.3 seconds). This rules out the insert statement itself as the bottleneck.

The real cost concentration is in two read queries that run as part of assembling a booking response:

- The driver active-ride check, 12.6ms average but spiking to 1.1 seconds, is the single largest consumer of total database time across the whole test, over 3 minutes of cumulative execution. It runs 14,590 times, far more often than there were actual booking attempts, suggesting it is called at multiple points in the flow, both when a driver goes online and again during matching.
- The nearby-driver and session lookup produced the single worst individual query execution recorded anywhere in the database during testing, 4.07 seconds against a 3.9ms average, an extreme, one-off spike rather than a pattern, but worth a closer look at what triggered it, likely lock contention with a concurrent write given how far it sits from its own average.

Revised recommendation for the engineering team: focus the `POST /api/v1/rides` investigation on the driver-availability and matching read path, specifically whether the active-ride check has a supporting index on `(driver_id, status)`, and whether it can be called fewer times per request, rather than the `rides` table INSERT itself, which is already fast.

9. Detailed Findings and Anomalies

9.1 Load-generation infrastructure sizing

Methodology note, not an application issue.

During test preparation, an initial attempt to run the full target concurrency (3,000 simulated riders) from a memory-constrained generation server resulted in that server running out of memory and being terminated by its own operating system before the test could complete, twice, at 46% and 54% of connection setup respectively. This was a load-generator sizing issue, not a staging environment issue: the target application was never actually under load when this occurred, since the test never got far enough to send real traffic. The fix was to right-size the generation server and reduce simulated rider count to 1,000 for this phase, which resolved the issue completely; both subsequent tests ran cleanly with substantial memory headroom to spare.

9.2 Driver go-online and create booking failures

Across both load tests, a small number of driver go-online calls (2 of 290, then 4 of 292) and, in the elevated test, a small number of create-booking calls (11 of 5,999) failed. In both tests this remained well under 1% of the relevant calls, and the failure rate did not worsen disproportionately to the 33x increase in load between the two tests, going from 0.7% to 1.4% is consistent with normal statistical variance at these small sample sizes rather than a load-driven degradation. Flagged as a minor item worth a closer look during a future test cycle, not a finding that affects the pass or fail verdict of this test.

9.3 Slow requests, 1 second or longer, full breakdown

Pulled directly from the application's own request logs (Grafana Cloud Loki) for both test windows, filtered to every request that took 1 second or longer to respond.

Baseline test, 14:38-14:59 IST: zero requests 1 second or longer. No slow requests at all at this load level.

Elevated test, 15:15-15:36 IST: 34 requests 1 second or longer, out of 24,410 total requests (0.14%).

Endpoint	Slow requests ($\geq 1s$)
POST /api/v1/rides (create booking)	30
POST /api/v1/rides/:id/cancel	3
POST /api/v1/pricing/estimate	1

Time window (IST)	Slow requests	Slowest in window
15:23:31 – 15:23:35 (4 seconds)	24	1,747ms
15:25:01 – 15:25:02	4	1,924ms

15:26:01	3	1,105ms
15:27:25	2	2,304ms and 2,214ms , the two slowest requests of the entire test, back to back in the same second

71% of all slow requests, 24 of 34, happened within one 4-second window. This shape, brief clustered spikes rather than a gradual slowdown, is the signature of short-lived contention rather than the endpoint being slow on average. It lines up with Section 9.2's create booking failures, which most likely occurred in or near these same windows.

9.4 Database behavior during the slow-request window

Metric	Baseline (Section 4.4)	During 15:23-15:24 IST spike
CPU utilization	~6-13%	43-53%
Active connections	3-14	up to 48
Write latency	~1-2ms	~1-2ms, unchanged
Disk queue depth	—	~0.15-0.2, low, unchanged

CPU and connection count both spiked sharply and specifically during the same window as the slow requests, while write latency and disk queue depth stayed flat. This rules out storage I/O as the cause and points at CPU-bound query execution or connection and lock contention during concurrent ride creation, consistent with several transactions briefly competing for the same rows rather than a disk bottleneck. Section 8 narrows this down further to the specific queries involved.

10. Conclusions and Recommendations

1. **The application functioned correctly under every load level tested**, with check pass rates of 99.92%, 99.94%, and 100% across the baseline, elevated, and spike tests respectively, and every defined latency and error-rate threshold passing in every case.
2. **Auto-scaling works correctly in both directions**, verified with precise timestamps: capacity added within 2 minutes 22 seconds of demand crossing the trigger, capacity removed within 15 minutes 11 seconds of demand subsiding, zero dropped connections during either transition.
3. **The system absorbed a sudden connection storm cleanly at every scale tried**, from 500 up to 2,000 concurrent real-time connections established within a 10-second surge, zero connection failures at any level.
4. **No infrastructure failures, crashes, or data issues occurred** at any load level, in any test.
5. **One concrete optimization target identified and narrowed to two specific queries:** ride creation under concurrent write pressure. The booking INSERT itself is fast (1.8ms average); the cost sits in a driver-availability check that is the largest single consumer of database time across the whole test, and a driver-matching lookup that produced the single worst query execution recorded (4.07 seconds). Scoped and actionable, the endpoint's overall response time stayed healthy throughout.
6. **Database health is strong independent of today's findings:** zero deadlocks recorded all-time, a 4-in-532,615 rollback rate, and a 99.67% cache hit ratio.
7. **Two configuration gaps in the spike-test script were found, fixed, and re-validated** within the same test cycle, both traced to specific evidence (idle database connection pool, no errors logged, healthy response times throughout) rather than assumed. Both are now fixed for every future run of this test.
8. **This test phase deliberately did not yet exercise the full target concurrency** of 6,000 riders and 400 drivers specified in the original production-readiness plan. That remains the next phase, once driver test-account count is increased and client scheduling is confirmed.

Report compiled from live AWS CloudWatch infrastructure metrics, k6 test-run output, staging database monitoring data, and PostgreSQL query statistics, captured during the test windows listed above. Source data was recorded in UTC; all timestamps in this report are converted to IST (UTC+5:30).